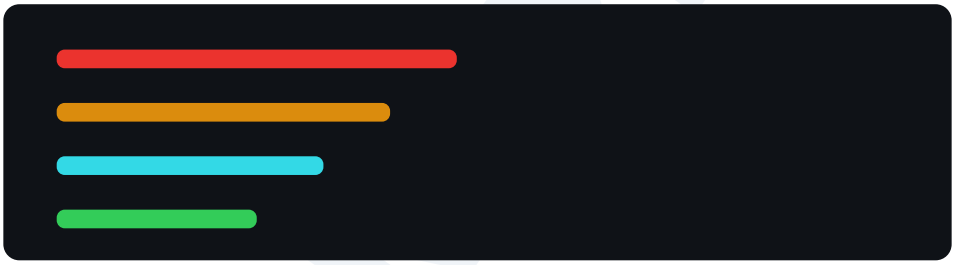
Sample Flight Crew Operations Manual

EICAS Ref sample library for the 777



Fictional content for demonstration only.

Not a real manual. Never use it for flight.

SAMPLE

Air Systems Chapter 2

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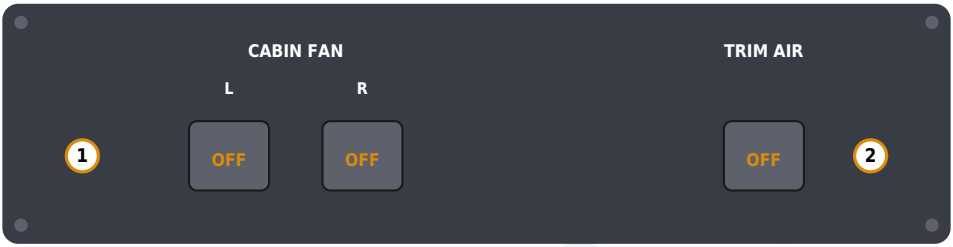
Air Systems Description 2.20
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Duct Temperature Control 2.20.2

EICAS Messages 2.30
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SAMPLE

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Panel Overview



1 CABIN FAN Switches

ON (in) - the sample fan runs.
OFF (out) - the fan stops and the OFF light shows.

2 TRIM AIR Switch

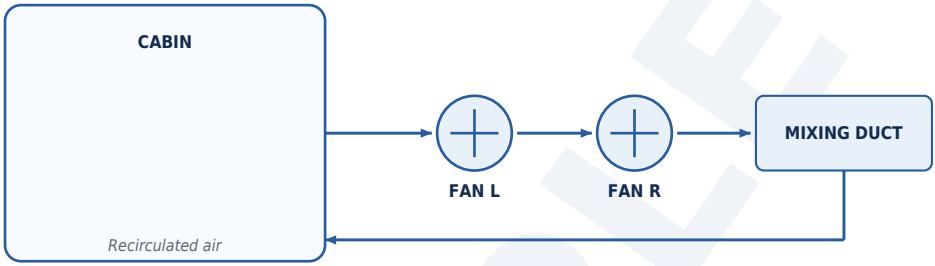
ON (in) - trim air is added to each duct.
OFF (out) - the trim air valve closes.

SAMPLE

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Cabin Fan System

Two sample fans draw air from the cabin and return it to the mixing duct, where it joins fresh air from the packs. Either fan keeps the cabin comfortable on its own.

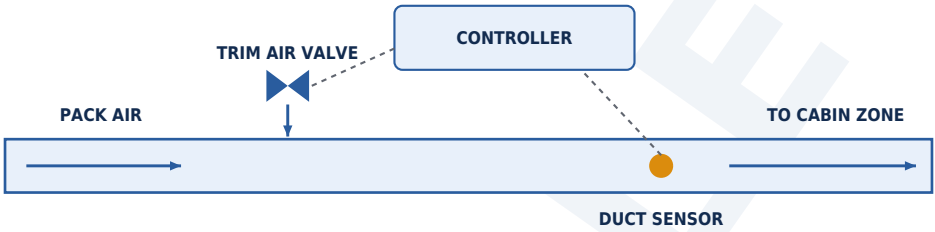


When a fan fails, the CABIN FAN L or R message shows and the fan's OFF light comes on.

SAMPLE

Duct Temperature Control

A sample controller trims the air in each duct to the temperature selected for its zone. It adds hot trim air through a valve and protects the duct from overheating.



If the duct becomes too hot, the controller closes the trim air valve and the DUCT TEMP HIGH message shows.

SAMPLE

Air Systems Chapter 2

EICAS Messages Section 30

Message	Level	Aural	Message Logic
CABIN FAN L, R	Advisory		A cabin fan has failed.
DUCT TEMP HIGH	Caution	Beeper	Duct temperature is high.

SAMPLE

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Anti-Ice, Rain Chapter 3

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Anti-Ice, Rain EICAS Messages 3.30.1

SAMPLE

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Panel Overview



1 WING ANTI-ICE Selector

AUTO - the sample valves open when ice is detected.
ON - the valves are open.
OFF - the valves are closed.

2 ICE DETECTOR Light

Shows when the sample detector senses ice.

SAMPLE

Wing Anti-Icing

Warm sample bleed air flows through two valves to the leading edge of each wing. The valves open and close with the WING ANTI-ICE selector.
The ANTI-ICE WING message shows when a valve position disagrees with the selector.

SAMPLE

Ice Detection

A sample ice detector on the forward fuselage tells the system when ice forms, so AUTO can open the valves without crew action.
The ICE DETECTOR message shows when the detector has failed. Select the wing anti-ice manually.

SAMPLE

Anti-Ice, Rain Chapter 3

EICAS Messages Section 30

Message	Level	Aural	Message Logic
ANTI-ICE WING	Caution		A wing anti-ice valve disagrees with its selection.
ICE DETECTOR	Status		The sample ice detector has failed.

SAMPLE

Electrical Chapter 6

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SAMPLE

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Panel Overview



1 GALLEY Switch

ON (in) - the sample galleys are powered.
OFF (out) - galley power is removed.

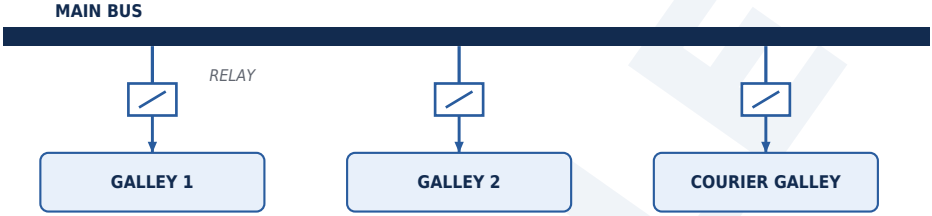
2 BATTERY Switch

ON (in) - the battery can power the standby buses.
OFF (out) - the battery is disconnected.

SAMPLE

Galley Power

The sample galleys are powered from the main bus through relays that open automatically when the electrical load is high, and close again when it falls.

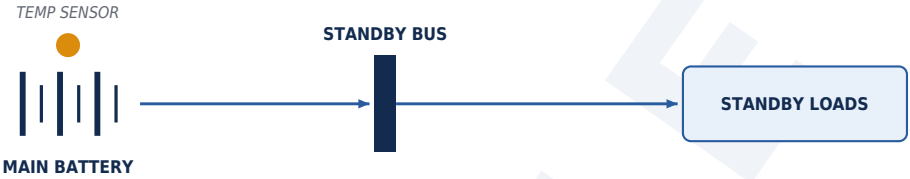


The GALLEY POWER message shows when a galley relay stays open.

SAMPLE

Main Battery

The sample battery powers the standby buses when no other source is available. A sensor watches its temperature while it charges.



The BATTERY TEMP message shows when the battery is too warm.

SAMPLE

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Electrical Chapter 6

EICAS Messages Section 30

Message	Level	Aural	Message Logic
GALLEY POWER	Advisory		Galley power is lost.
BATTERY TEMP	Caution	Beeper	Battery temperature is high.

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Engines, APU Chapter 7

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Engines, APU EICAS Messages 7.30.1

SAMPLE

Panel Overview



1 APU Selector

OFF - the sample APU stops after a cooldown.
ON - the APU runs and supplies air and power.
START - momentary; starts the APU.

2 FAULT Light

Shows with an APU fault or automatic shutdown.

SAMPLE

APU Operation

The sample auxiliary power unit supplies electrical power and bleed air on the ground and in flight. Its inlet door opens before the start.
The APU LIMIT message shows when the APU exceeds a sample limit and protects itself.

SAMPLE

APU Shutdown

The APU shuts down automatically after some sample faults. A cooldown runs before a normal shutdown.
The APU SHUTDOWN and APU FAULT messages show when the APU has stopped by itself.

SAMPLE

Engines, APU Chapter 7

EICAS Messages Section 30

Message	Level	Aural	Message Logic
APU FAULT	Advisory		A sample APU fault is detected.
APU LIMIT	Advisory		The APU has exceeded a sample limit.
APU SHUTDOWN	Advisory		The APU has shut down automatically.
APU DOOR	Status		The APU inlet door disagrees with its command.

SAMPLE

Flight Controls Chapter 9

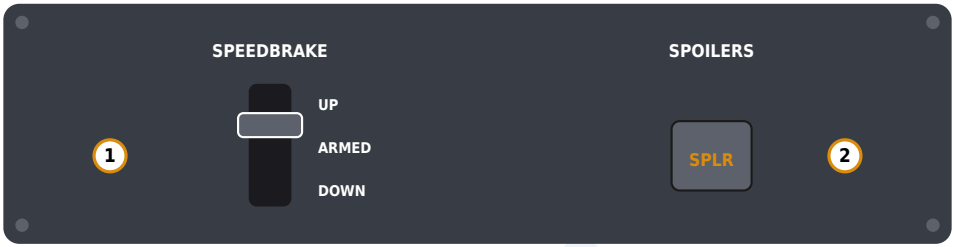
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Flight Controls EICAS Messages 9.30.1

SAMPLE

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Panel Overview



1 Speedbrake Lever

DOWN - all sample spoiler panels are retracted.
ARMED - the panels extend on landing.
UP - the panels extend fully.

2 SPOILERS Light

Shows with a sample spoiler fault.

SAMPLE

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Spoilers

Fourteen sample spoiler panels, seven on each wing, help the ailerons with roll control and reduce lift after landing.
Each panel has its own actuator. The SPOILERS message shows when one or more panels have a fault.

SAMPLE

Speedbrakes

The sample speedbrake lever extends the spoiler panels together in flight and on the ground. When ARMED, they extend automatically on landing.
The SPEEDBRAKE EXTENDED message shows when the speedbrakes are extended at low altitude.

SAMPLE

Flight Controls Chapter 9

EICAS Messages Section 30

Message	Level	Aural	Message Logic
SPOILERS	Advisory		A sample spoiler fault is detected.
SPOILER ACTUATOR	Status		A spoiler actuator has failed.
SPEEDBRAKE EXTENDED	Caution	Beeper	Speedbrakes extended at low altitude.

SAMPLE

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Fuel Chapter 12

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SAMPLE

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Panel Overview



1 AUX PUMP Switch

ON (in) - the auxiliary sample pump runs.
OFF (out) - the pump stops. PRESS shows when the pump is ON with low pressure.

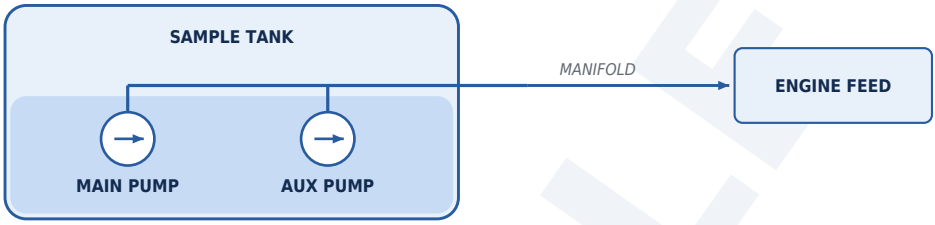
2 Fuel Temperature Indication

Shows the sample tank fuel temperature.

SAMPLE

Auxiliary Fuel Pump

An auxiliary sample pump supports the main pumps during takeoff and climb. It feeds the same manifold as the main pumps.

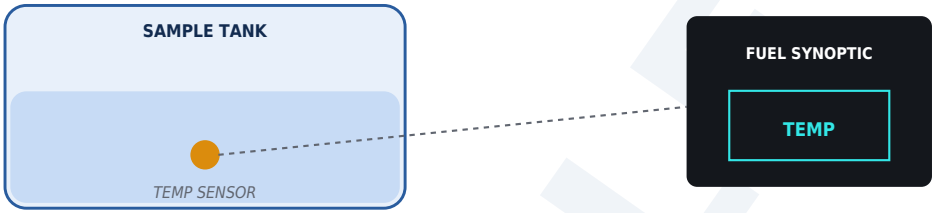


The FUEL PUMP AUX message shows when the pump is on and its output pressure is low.

SAMPLE

Fuel Temperature

A sensor in the sample tank shows the fuel temperature on the fuel synoptic. Long flights in cold air slowly cool the fuel.



The FUEL TEMP LOW message shows when the fuel reaches the sample limit.

SAMPLE

Fuel Chapter 12

EICAS Messages Section 30

Message	Level	Aural	Message Logic
FUEL PUMP AUX	Advisory		Auxiliary pump pressure is low.
FUEL TEMP LOW	Caution	Beeper	Fuel temperature is low.

SAMPLE

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Hydraulics Chapter 13

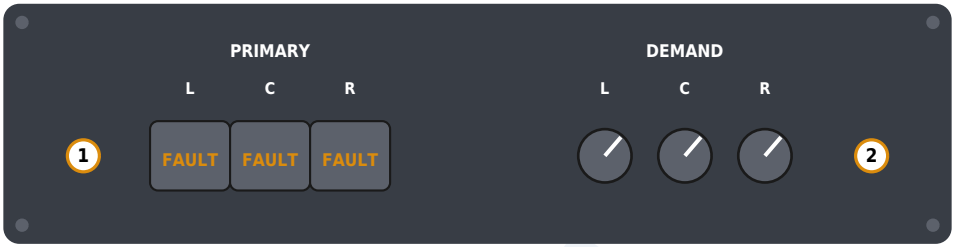
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Hydraulics EICAS Messages 13.30.1

SAMPLE

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Panel Overview



1 PRIMARY Pump Switches

ON (in) - the sample engine-driven pump runs.
OFF (out) - the pump stops and FAULT may show.

2 DEMAND Pump Selectors

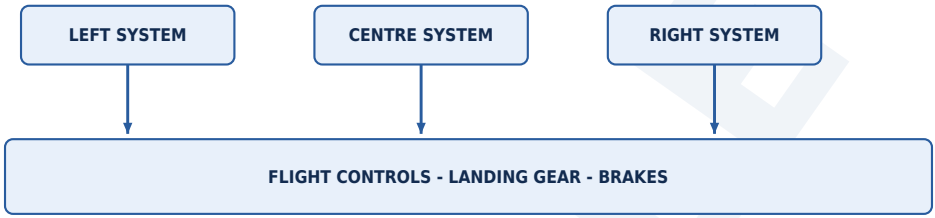
OFF - the sample demand pump is off.
AUTO - runs when the sample demand is high.
ON - runs all the time.

SAMPLE

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Hydraulic Systems

Three sample hydraulic systems, left, centre and right, power the flight controls, the landing gear and the brakes. Each system has its own reservoir and pumps.

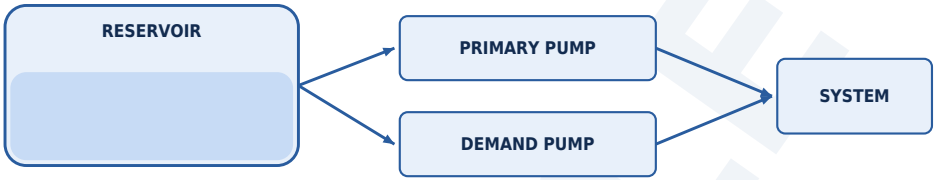


The HYD PRESS SYS L, C or R message shows when a system's pressure is low.

SAMPLE

Demand Pumps

Each sample system has a demand pump that supports the primary pump when demand is high, such as while the gear and flaps move.



The HYD PRESS DEM message shows when a demand pump that should run has low pressure.

SAMPLE

Hydraulics Chapter 13

EICAS Messages Section 30

Message	Level	Aural	Message Logic
HYD PRESS SYS L, C, R	Caution	Beeper	System pressure is low.
HYD PRESS DEM R	Advisory		Demand pump pressure is low.
HYD QTY LOW R	Advisory		Reservoir quantity is low.

SAMPLE

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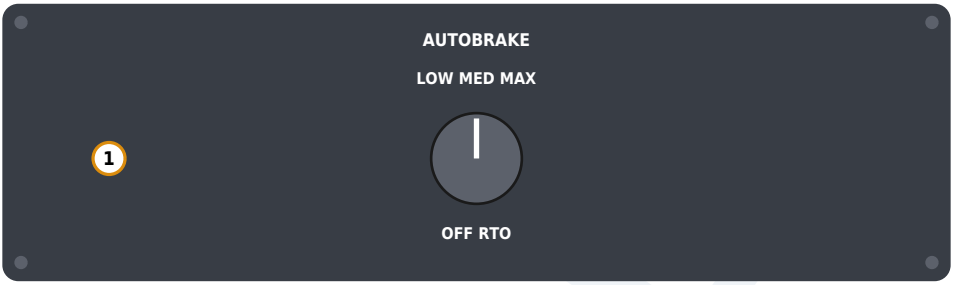
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SAMPLE

Panel Overview



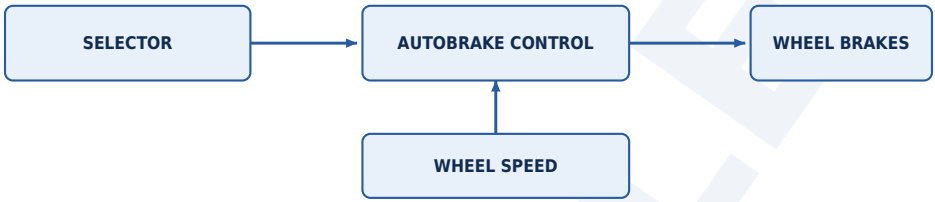
1 AUTOBRAKE Selector

OFF - the autobrake is off.
LOW, MED, MAX - the sample deceleration rate for landing.
RTO - maximum braking for a rejected takeoff.

SAMPLE

Autobrake System

The sample autobrake applies the brakes after landing at the rate set on the selector, and releases them when the pilot brakes manually.



When the system finds a fault it disarms, the selector should be set OFF and the AUTOBRAKE message shows.

SAMPLE

Manual Braking

The brake pedals control each wheel brake directly. Manual braking is always available, with or without the autobrake.



Use the sample landing distance for manual braking when the autobrake is not used.

SAMPLE

Landing Gear Chapter 14

EICAS Messages Section 30

Message	Level	Aural	Message Logic
AUTOBRAKE	Advisory		Disarmed or has a fault.

SAMPLE

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